

Dylan Folsom

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Education

Princeton University

Princeton, NJ

Ph.D. in Physics

2027, est.

Dissertation *Computational Models for Dark Matter in the Milky Way and its Satellites* advised by M. Lisanti

Vanderbilt University

Nashville, TN

B.A. *cum laude* with Highest Honors in Physics

2020

Majors in Physics, Mathematics, and Classical Studies; minor in Computer Science

Teaching, Mentorship, & Service

Prison Teaching Initiative

Princeton, NJ

Course Coordinator & Instructor

Fall 2026

Facilitated syllabus creation, instructional team meetings, and course logistics for MATH 101 and MATH 070 at Garden State Youth Correctional Facility, where I also volunteered as an instructor

Coding Foundations of Research Assistant

Summer 2026

Led interns in the Scientific Thought and Practice course to discover the practices of scientific inquiry, to design and conduct experiments, and to develop skills of observation and critical thinking

Princeton University Department of Physics

Princeton, NJ

Teaching Assistant – Quantum Mechanics sequence

2026

Helped develop new curricula for the Quantum Mechanics sequence, incorporating computational approaches to problem-solving and developing students' Python skills

- PHY 207 (From Classical to Quantum Mechanics) Fall 2026

Teaching Assistant – Introductory Physics

2021–2025

Worked closely with the lab director to emphasize structured quantitative inquiry in the first-year physics courses

- PHY 105 Lab (Advanced Physics, Mechanics) Fall 2023, 2024, 2025
- PHY 104 Lab (General Physics II, Electricity and Magnetism) Spring 2023
- Additionally assisted with PHY 115 (Fall 2022), 110 (Summer 2022), 109 (Spring 2022), & 105 (Fall 2021)

Undergraduate Mentor

2024–2026

- Tal Shpigel – Advised during his OURSIP internship project, *Stellar Tracers of Dark Matter Kinematics in GSE-like Mergers from the IllustrisTNG Simulation*, and further work, which led to *Astrophys. J.* **1003**, 102 (2026)

Butler College

Princeton, NJ

Resident Graduate Student

2021–2022

Coordinated community-building activities for the residential college; mentored a group of first-year students

Publications

- [8] **DF**, C. Blanco, M. Lisanti, M. Vogelsberger, *Ubiquitous Corotation of Dark Matter Halos: Implications for Direct Detection* [2608.00161]
- [7] K. Raman, **DF**, M. Kaplinghat, M. Lisanti, B. Safdi, *Semi-analytic Inference of Cold Dark Matter Satellite Densities, I: Comparison to Ultra-faint Dwarf Kinematics* [2607.27316]
- [6] K. Raman, **DF**, M. Kaplinghat, M. Lisanti, B. Safdi, *Semi-analytic Inference of Cold Dark Matter Satellite Densities, II: Implications for Dark Matter Indirect Detection Constraints* [2607.27326]

- [5] L. Necib, **DF**, E. Davies, N. Starkman, A. Thoyas, *Galactic Amnesia: The Information Washout of the Milky Way Merger History* [2605.04138]
- [4] T. Shpigel, **DF**, M. Lisanti, L. Necib, M. Vogelsberger, L. Hernquist, *How to Build an Empirical Speed Distribution for Dark Matter in the Solar Neighborhood*, *Astrophys. J.* **1003**, 102 (2026) [2510.21914]
- [3] **DF**, C. Blanco, M. Lisanti, L. Necib, M. Vogelsberger, L. Hernquist, *Dark Matter Velocity Distributions for Direct Detection: Astrophysical Uncertainties are Smaller Than They Appear*, *Phys. Rev. Lett.* **135**, 211004 (2025) [2505.07924]
- [2] **DF**, M. Lisanti, L. Necib, D. Horta, M. Vogelsberger, L. Hernquist, *Cosmological Simulations of Stellar Halos with Gaia Sausage–Enceladus Analogues: Two Sausages, One Bun?*, *Astrophys. J.* **983**, 119 (2025) [2408.02723]
- [1] **DF**, O. Slone, M. Lisanti, F. Jiang, M. Kaplinghat, *Probabilistic Inference of the Structure and Orbit of Milky Way Satellites with Semi-Analytic Modeling*, *Mon. Not. R. Astron. Soc.* **536**, 2891 (2025) [2311.05676]

Talks

Ubiquitous Corotation of Dark Matter Halos: Implications for Direct Detection Phenomenology Symposium – PHENO	<i>University of Pittsburgh</i> May 2026
The Dark Matter Around the Milky Way TIPOTS Graduate Student Colloquium	<i>Princeton University</i> May 2026
Diversity in Ultra-faint Dwarf Densities and Implications for Indirect Detection Particle Astrophysics and Cosmology Meeting Around NYC – PACMAN	<i>Flatiron Institute</i> January 2026
Halo-to-Halo Similarity in IllustrisTNG Dark Matter Velocity Distributions Dark Matter Symposium – UCLA DM (poster)	<i>University of California, L.A.</i> March 2025
Probabilistic Inference of the Structure and Orbit of Milky Way Satellites with Semi-Analytic Modeling TeV Particle Astrophysics – TeVPA	<i>University of Chicago</i> August 2024
Two Sausages, One Bun? GSEs in TNG Small Galaxies, Cosmic Questions II (poster)	<i>Durham University</i> July 2024
Probabilistic Inference of the Structure and Orbit of Milky Way Satellites with Semi-Analytic Modeling Local Local Group Group – L2G2	<i>Columbia University</i> December 2023
Semi-Analytic Inference of the Dark Matter Structure of Satellite Galaxies Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond	<i>Flatiron Institute</i> June 2023

Awards & Honors

2025	Kusaka Memorial Prize in Physics	<i>Princeton University Department of Physics</i>
2024	Joseph H. Taylor Graduate Student Fellowship	<i>Princeton University Department of Physics</i>
2020 & 2021	President’s Fellowship	<i>Princeton University Graduate School</i>
2020	Joseph Henry Merit Prize	<i>Princeton University Department of Physics</i>
2016–2020	Nancy and William McMinn Scholarship for Physics	<i>Vanderbilt University Department of Physics</i>
2016–2020	STEM and Business Scholarship	<i>Scarlett Family Foundation</i>
Inducted 2019	Pi Mu Epsilon Mathematics Honors Society	
Inducted 2018	Sigma Pi Sigma Physics Honors Society	